ICT2132 · Object Oriented Programming Practicum

Core Concepts: Swing Events and Validation

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Swing uses an event-dispatch thread for UI updates. Listeners should remain responsive and avoid long blocking work. Validate input at the boundary, then enforce important rules again in the service layer.

2. Example and reasoning

A save button can read a title and offering, show a validation message for blank input, then call a service. A service test should still reject a blank title even when the UI already checks it.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “core concepts: swing events and validation” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.